

VLE Simulation Guided Inquiry

Understanding Saturated vs. Superheated Vapor

System Specs: Constant Pressure ($P = 1 \text{ atm}$) | **Feed Composition:** 20 mol% H_2O , 80 mol% Air

Introduction & Objectives

This guided inquiry uses the interactive simulation to explore the phase behavior of a condensable component (water) mixed with a non-condensable gas (air). By the end of this activity, you will be able to:

- Define and physically distinguish between a **saturated vapor** and a **superheated vapor**.
- Explain why water can evaporate or exist entirely as a superheated vapor far below its normal boiling point of 100°C in a mixture.
- Calculate the partial pressure and dew point behavior of water in air.

1 Part 1: Key Definitions

In thermodynamics, the state of a vapor depends heavily on whether it is in equilibrium with its liquid phase. Let us define the two main states you will observe in the cylinder.

1. Saturated Vapor

- **What it is:** A vapor that exists in direct thermodynamic equilibrium with its liquid phase. The gas phase is “holding” the maximum possible amount of the condensable component for its current temperature (it is at its saturation limit). Any attempt to compress the system or lower the temperature will result in immediate condensation of the vapor into liquid.
- **In the Simulation:**
 1. Set the temperature slider to 40°C .
 2. Observe the liquid water layer at the bottom of the cylinder. The gas space above it is a **saturated mixture**.
 3. Look at the data readout. The vapor phase composition of water ($y_{\text{H}_2\text{O}}$) is low (around 7.3 mol%). Because the system started with 20 mol% water overall, the remaining 12.7 mol% is forced to remain as a subcooled liquid at the bottom.

2. Superheated Vapor

- **What it is:** A vapor at a temperature higher than its saturation temperature (or dew point) at a given pressure. It is not in contact with any liquid phase, meaning the gas phase is “unsaturated” and has the capacity to hold more vapor if liquid were introduced.
- **In the Simulation:**
 1. Move the temperature slider up to 80°C .

2. Observe that all liquid has evaporated, leaving only gas in the cylinder. The piston has adjusted upwards to maintain constant pressure.
3. Look at the data readout. The water composition in the vapor phase ($y_{\text{H}_2\text{O}}$) is now fixed at exactly 20 **mol%**. This represents the entire inventory of water in the closed system.

2 Part 2: The 100°C Myth vs. The Dew Point

A common misconception is that liquid water cannot exist below 100°C at 1 atm, or conversely, that water must reach 100°C to become a superheated vapor.

The Explanation

- **Normal Boiling Point (100°C):** This is the temperature at which the saturation vapor pressure of **pure water** ($P_{\text{H}_2\text{O}}^*$) equals the total ambient pressure (1 atm).
- **The Dew Point (60.4°C):** In our mixture, water is not pure; it constitutes only 20 mol% of the overall system. By Raoult's Law / Dalton's Law of partial pressures, the maximum partial pressure that water vapor can exert when all water is evaporated is:

$$P_{\text{H}_2\text{O}} = y_{\text{H}_2\text{O}} \cdot P_{\text{total}} = 0.20 \cdot 1.0 \text{ atm} = 0.20 \text{ atm}$$

The **Dew Point** is the specific temperature where water's saturation vapor pressure ($P_{\text{H}_2\text{O}}^*$) is exactly equal to this required partial pressure of 0.20 atm. Using vapor pressure data (such as the Antoine Equation), this threshold occurs at precisely 60.4°C.

Thermodynamic Thresholds:

- **Below 60.4°C:** The vapor pressure of water is less than 0.20 atm ($P_{\text{H}_2\text{O}}^* < 0.20 \text{ atm}$). Thus, the gas phase cannot hold all the water. Liquid water must condense out until the partial pressure of water in the remaining gas equals $P_{\text{H}_2\text{O}}^*$. The gas is **saturated**.
- **Above 60.4°C:** The vapor pressure of water is greater than 0.20 atm ($P_{\text{H}_2\text{O}}^* > 0.20 \text{ atm}$). The gas phase is capable of holding more than 20 mol% water. Because all available water has evaporated, no liquid remains, and the vapor becomes **superheated**.

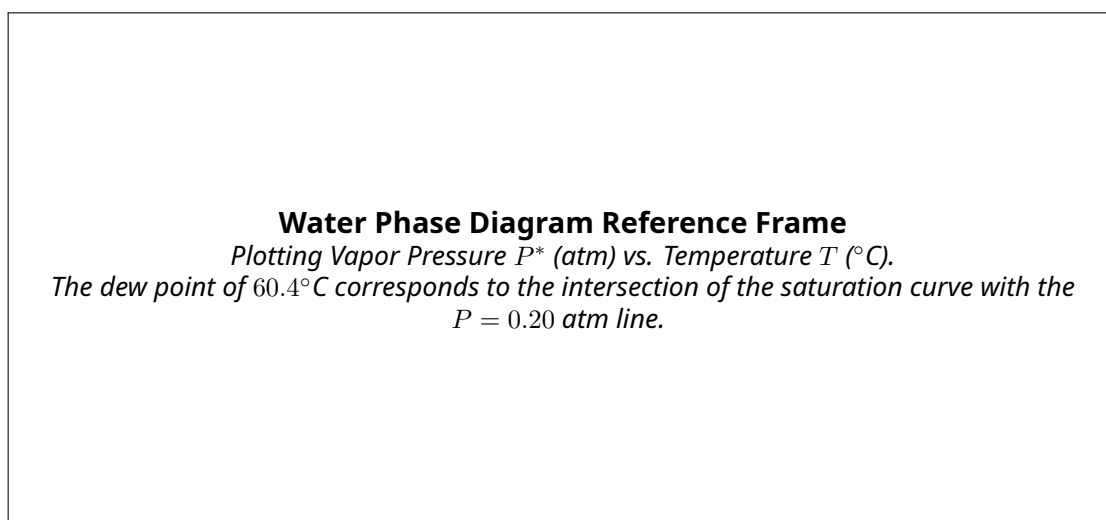


Figure 1: A conceptual representation of the water vapor pressure curve.

3 Check for Understanding

Answer the following questions using the simulation to verify your understanding.

Guided Questions

1. Set the slider to 25°C.

- Is the vapor phase saturated or superheated? _____
- What is the current mole % of H₂O in the vapor? _____

2. Slowly transition the slider from 25°C to 60°C.

- Describe what happens to the volume of the liquid phase: _____
- Describe what happens to the mole % of H₂O in the gas: _____

3. Position the slider exactly above the dew point mark (e.g., 65°C).

- Is the vapor saturated or superheated? _____
- Why is there no liquid water remaining in the cylinder? _____

4. Advance the slider further to 105°C.

- Does the mole % of water in the gas change as you heat the system past 60.4°C? Why or why not?

Guided Inquiry Solution Key

1. At 25°C:

- The vapor is **saturated**. Liquid is present.
- The vapor contains roughly 3.1 **mol% H₂O**.

2. From 25°C to 60°C:

- The liquid phase volume **decreases** as liquid water evaporates.
- The mole % of H₂O in the gas phase **increases** (climbing up towards 20%).

3. At 65°C (Above Dew Point):

- The vapor is **superheated**.
- No liquid remains because the air's holding capacity for water vapor at 65°C (saturated limit ≈ 25 mol%) is greater than the total amount of water in the system (20 mol%).

4. At 105°C:

- No, the composition remains locked at 20 **mol%**. This is because the system is closed and all water is already in the gas phase; there is no remaining liquid reservoir to evaporate and increase the gas-phase concentration.